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Homework Project Technical Report

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1 Part I

1.1 Gearbox Analysis

Group	$P_{in}(kW)$	$n_{in}(rpm)$
23	28	1000

Power is provided to the gearbox through gear G1 which meshes with gear G2. The gear G1 rotates counterclockwise seen from the x,y-plane. All the required parameters are defined as follows:

	Gear 1	Gear 2	Gear 3	Gear 4
normal module, $m_n(mm)$	3	3	4	4
number of teeth, z	16	48	21	63
helix angle, ψ (degrees)	15	15	15	15
normal pressure angle, ϕ_n (degrees)	20	20	20	20
transmission accuracy level, Q_v	8	8	8	8
tooth face width (mm)	45	45	56	56

Efficiency : $\eta = 1$

For speed ratios, we use: $i_{ij} = \frac{n_i}{n_j} = \frac{Z_{G_j}}{Z_{G_i}}$

For gear radii, since we have helical gears, we use the formula: $r = \frac{m_n \cdot Z_i}{2 \cdot \cos(\psi)}$

Then we find the initial torque by using the initial power with initial angular speed. After this step, we calculate all the torques by using the angular speeds found from the speed ratios.

Speed ratio	Value
i_{12}	3
i_{34}	3
i_{14}	9

Gears radii	Value(mm)
G1	24.846
G2	74.539
G3	43.482
G4	130.445

Angular speed	Value(rad/s)
ω_1	104.72
ω_2	34.9066
ω_3	34.9066
ω_4	11.6355

Given the input power P and speed n_1 (or ω_1 if expressed in rad/s), the torque C_1 gear G_1 exerts is:

$$C_1 = P n_1 \frac{2\pi}{60} = 267.38 Nm$$

From the equilibrium of gears G_1 and G_2 , as shown in Figure 1, it is possible to establish all the components of the force transmitted between them:

$$C_1 \omega_1 = C_2 \omega_2 \rightarrow C_2 = 802.141 Nm$$

$$F_{t,21} = F_{t,12} = \frac{C_1}{r_1} = 10761.2 N$$

$$F_{r,21} = F_{r,12} = F_{t,21} \tan \phi_t = F_{t,21} \frac{\tan \phi_n}{\cos \psi} = 4054.94 N$$

$$F_{a,21} = F_{a,12} = F_{t,21} \tan \psi = 2883.46 N$$

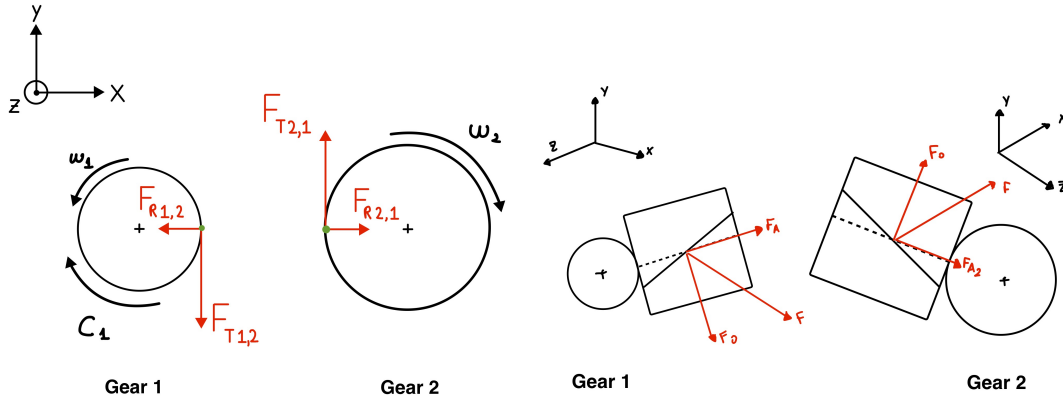


Figure 1 - Meshing of gear 1 and 2

Analogously, it is possible to find all the components of the force acting between helical gears G_2 and G_3 , starting by the fact that $C_2 = C_3$ since they are located on the same shaft:

$$C_4 = i_{34}C_3 = 2406.42Nm$$

$$F_{t,43} = F_{t,34} = \frac{C_3}{r_3} = 18447.8N$$

$$F_{r,43} = F_{r,34} = F_{t,43} \tan \phi_t = F_{t,43} \frac{\tan \phi_n}{\cos \psi} = 6951.32N$$

$$F_{a,43} = F_{a,34} = F_{t,43} \tan \psi = 4943.0796N$$

Putting all the values into tables we have:

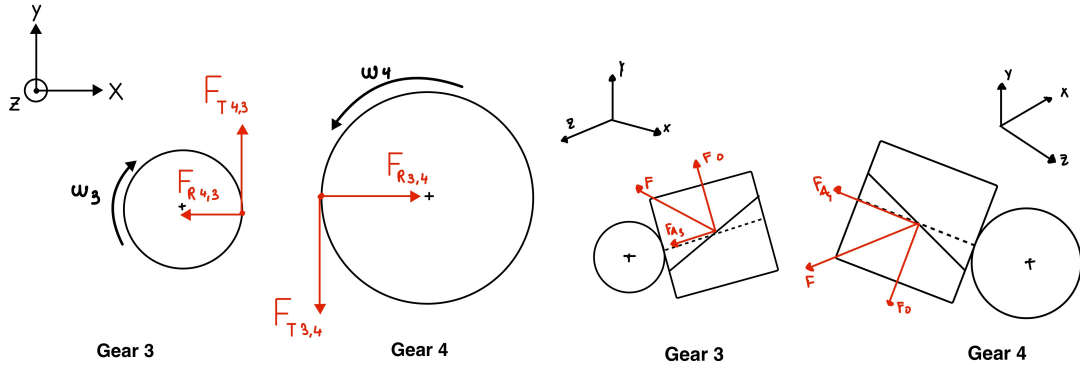


Figure 2 - Meshing of gear 3 and 4

F_t	Value (N)
$F_{t,21}$	10761.2
$F_{t,12}$	10761.2
$F_{t,43}$	18447.8
$F_{t,34}$	18447.8

F_r	Value (N)
$F_{r,21}$	4054.94
$F_{r,12}$	4054.94
$F_{r,43}$	6951.32
$F_{r,34}$	6951.32

F_a	Value (N)
$F_{a,1}$	2883.4631
$F_{a,2}$	2883.4631
$F_{a,3}$	4943.0796
$F_{a,4}$	4943.0796

By having all these values we can calculate $F_{out,r}$ and $F_{out,a}$:

$$F_{out,r} = \frac{C_4}{0.25} = 9625.69$$

$$F_{out,a} = F_{out,r} 0.25 = 2406.42$$

1.2 Reactions on bearings

Firstly, to find the reactions on each bearing, it is necessary to know the exact position where these reactions are applied. Considering shaft A1 we have:

$$a = 79 \text{ mm}$$

$$b = 70.5 \text{ mm}$$

$$c = 59.25 \text{ mm}$$

Since we know that the bearings on A, C, and E are cylindrical non-locating roller bearings, we know that they don't sustain any axial load. So $R_{z,A} = R_{z,C} = R_{z,E} = 0$. Below the pictures for shaft A₁ are presented with all the forces and reactions.

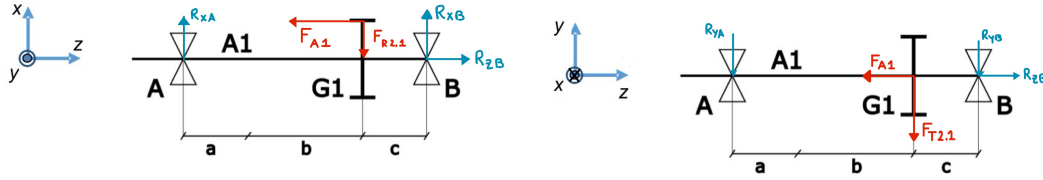


Figure 3 - Shaft A1 in the XZ and YZ plane

Let's start by writing down the equilibrium equations:

$$X) R_{xA} + R_{xB} - F_{r,21} = 0$$

$$Y) - R_{yA} - R_{yB} - F_{T,21} = 0$$

$$Z) - F_{A1} + R_{zB} + R_{zA} = 0$$

$$\curvearrowright A_Y) - F_{A1} \cdot r_1 + F_{R,21} \cdot (a + b) - R_{xB} \cdot (a + b + c) = 0$$

$$\curvearrowright A_X) - F_{T,21} \cdot (a + b) - R_{yB} \cdot (a + b + c) = 0$$

Following the same analogy for the shafts A2 and A3. Below are presented the figures of the two planes of each shaft with all the forces and the equilibrium equations.

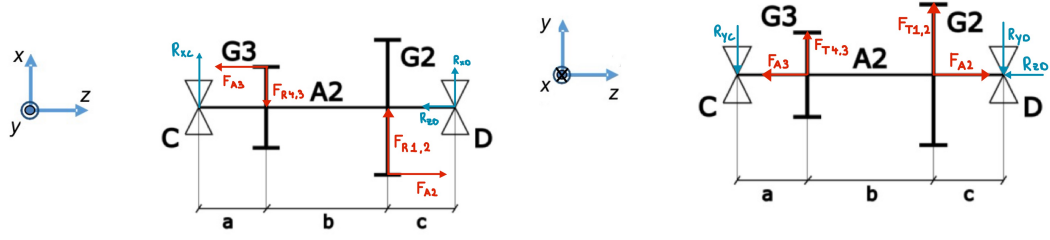


Figure 4 - Shaft A2 in XZ and YZ plane

Writing down the equilibrium equations for shaft A2:

$$X) R_{xC} + R_{xD} - F_{r,43} + F_{r,12} = 0$$

$$Y) - R_{yD} - R_{yC} + F_{T,12} + F_{T,43} = 0$$

$$Z) - F_{A3} + F_{A2} - R_{zD} = 0$$

$$\curvearrowright C_Y) F_{r,43} \cdot a - F_{A3} \cdot r_3 - F_{R,12} \cdot (a + b) - R_{xD} \cdot (a + b + c) = 0$$

$$\curvearrowright C_X) F_{T,43} \cdot a + F_{T,12} \cdot (a + b) - R_{yD} \cdot (a + b + c) = 0$$

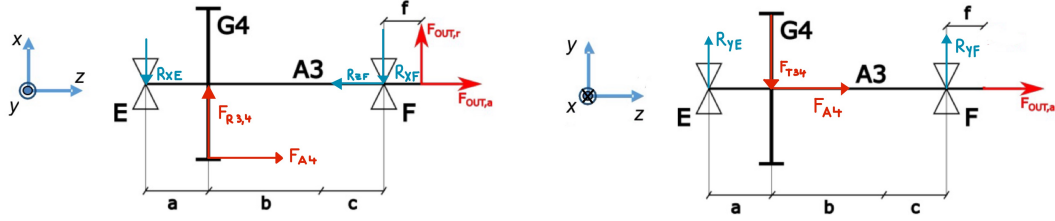


Figure 5 - Shaft A3 on XZ and YZ plane

Writing down equilibrium equations for shaft A3:

$$X) F_{r,34} - R_{xF} - R_{xE} + F_{out,r} = 0$$

$$Y) R_{yE} + R_{yF} - F_{T,34} = 0$$

$$Z) F_{out,a} + F_{A4} - R_{zF} = 0$$

$$\curvearrowright E_Y) - F_{r,34} \cdot a - F_{A4} \cdot r_4 + R_{xF} \cdot (a + b + c) - F_{out,r} \cdot (a + b + c + f) = 0$$

$$\curvearrowright E_X) - F_{T,34} \cdot a + R_{yF} \cdot (a + b + c) = 0$$

Solving the linear equations, we put the values in the table below:

Reactions	Value (N)	Reactions	Value (N)	Reaction	Value (N)
R_{xA}	1494.13	R_{xC}	5228.96	R_{xE}	-2457.1047
R_{xB}	2560.81	R_{xD}	-2332.57	R_{xF}	19034.115
R_{yA}	-3054.39	R_{yC}	14520.758	R_{yE}	11466.372
R_{yB}	-7706.85	R_{yD}	14688.298	R_{yF}	6981.4521
R_{zA}	0	R_{zC}	0	R_{zE}	0
R_{zB}	-2883.46	R_{zD}	-2059.6165	R_{zF}	7349.5024

Reactions on shaft A1

Now we also calculate the circumferential force on the bearings :

$$R_A = \sqrt{R_{xA}^2 + R_{yA}^2} = 3400.25N$$

$$R_B = \sqrt{R_{xB}^2 + R_{yB}^2} = 8121.16N$$

$$R_C = \sqrt{R_{xC}^2 + R_{yC}^2} = 15433.5N$$

$$R_D = \sqrt{R_{xD}^2 + R_{yD}^2} = 14872.4N$$

$$R_E = \sqrt{R_{xE}^2 + R_{yE}^2} = 11726.7N$$

$$R_F = \sqrt{R_{xF}^2 + R_{yF}^2} = 20274.1N$$

1.3 Internal actions and static verification

1.3.1 Internal actions

The internal actions of shaft A2 are required as a natural step toward the calculation of stresses and the static verification of the shaft. According to the data provided in the technical drawing, the dimensions between the four proposed sections can be obtained.

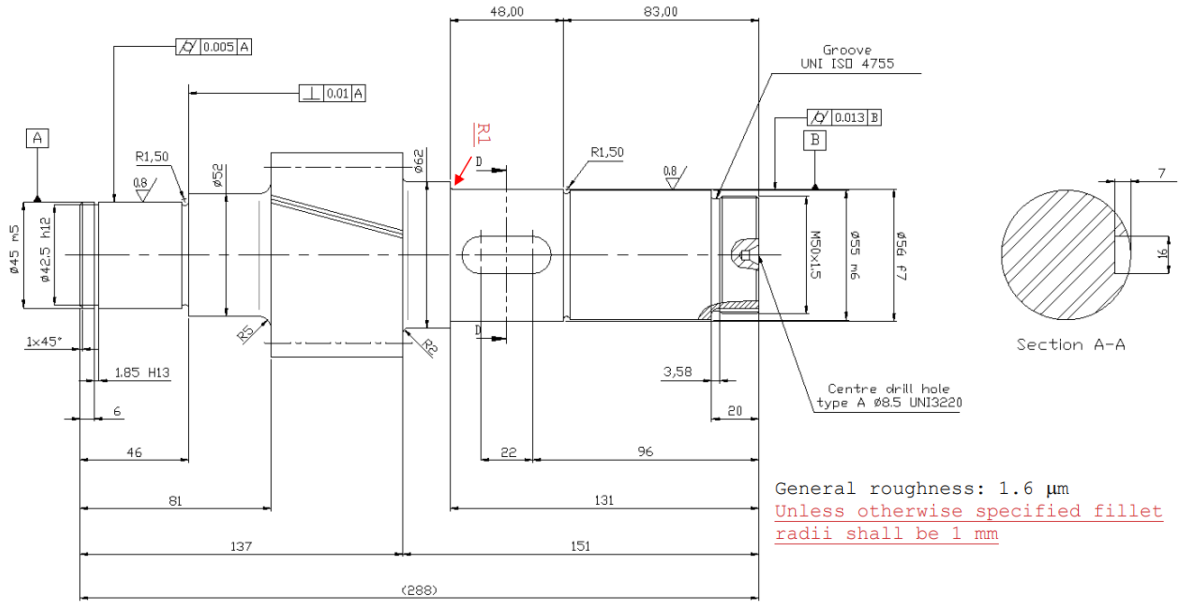


Figure 6 - Shaft A2 technical drawing

$$l_1 = 137 - 28 - 79 = 30\text{cm} \text{ (distance between the start of the shaft to the middle of the bearing)}$$

$$l_2 = a = 79\text{cm} \text{ (distance between the middle point of the bearing C to the Section V1)}$$

$$l_3 = (137 - 81)/2 = 28\text{cm} \text{ (distance between the section V1 to V2)}$$

$$l_4 = 151 - 131 = 20\text{cm} \text{ (distance between the section V2 to V3)}$$

$$l_5 = 45/2 = 22.5\text{cm} \text{ (distance between the section V3 to V4 by dividing the width of gear 2 in half)}$$

$$l_6 = c = 59.25\text{cm} \text{ (distance between the section V4 to the middle point of bearing D)}$$

$$l_7 = 288 - l_1 - l_2 - l_3 - l_4 - l_5 - l_6 = 49.25\text{cm} \text{ (distance between the middle point of bearing D until the end of the shaft)}$$

It is important to mention that for each internal action, the positive direction chosen is the one given by the reference frame of the shaft, which means that all moments are counterclockwise concerning their axes. Below, an image of the shaft in 3D with the reference frame and with the forces and the sections in it, is shown.

The table with all the results of the internal actions is presented below and it also includes the total bending moment $M_b = \sqrt{M_x^2 + M_y^2}$. To have a better idea of the internal actions throughout the whole shaft, the internal actions at the bearing's pressure points are included

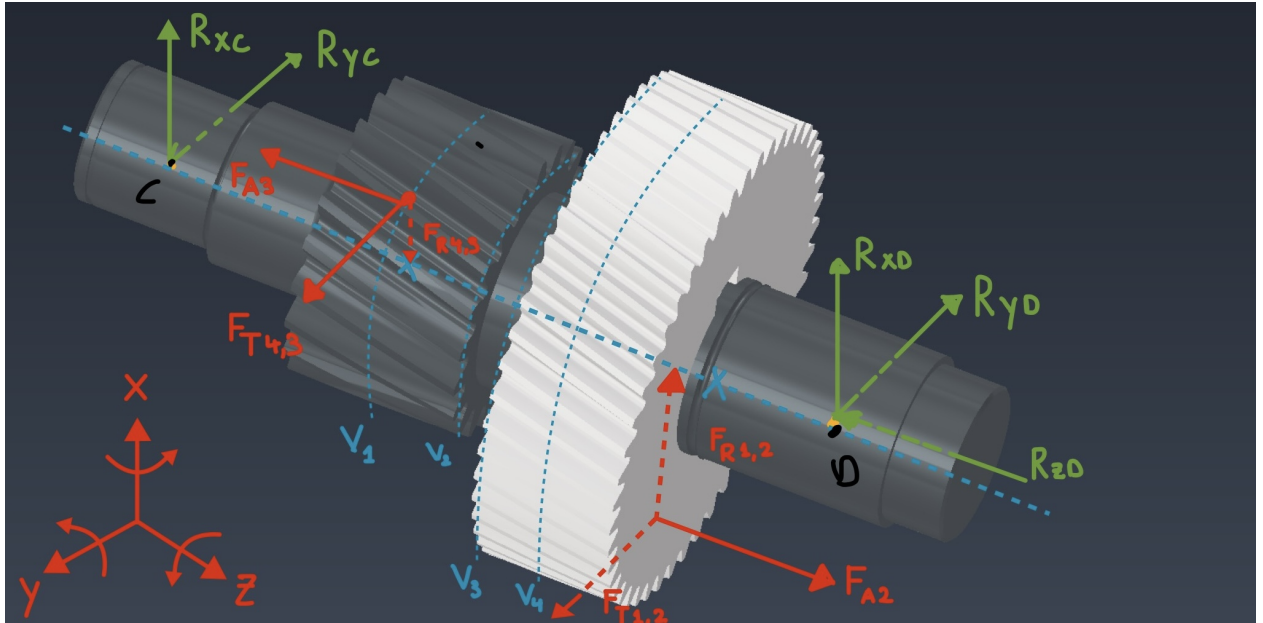


Figure 7 - Shaft A2 with the forces on each gear

Section	N (N)	T_y (N)	T_x (N)	M_x (Nm)	M_y (Nm)	M_b (Nm)	M_z (Nm)
C	0	-5228.9568	14520.758	0	0	0	0
V1	4943.0796	-3927.0668	1722.3626	1147.1398	198.15458	1164.1285	-802.14091
V2	4943.0796	-3927.0668	1722.3626	1037.182	149.92843	1047.9623	-802.14091
V3	4943.0796	-3927.0668	1722.3626	958.64064	115.48117	965.57122	-802.14091
V4	2059.62	-2332.57	-14688.3	870.282	-138.205	881.187	0
D	4119.23	-29376.6	0	0	0	0	0

The trends of each internal action are plotted in the graphs below. The 0 coordinate in the graphs refers to the starting point of the shaft. Then the point with coordinate 0.03 is the bearing pressure point C.

From that come sections V1, V2, V3, V4, bearing pressure point D, and the end of the shaft. We can spot that the bending moments around x and y and the total bending moment reach their peak in section V1 which is in the middle of gear 3.

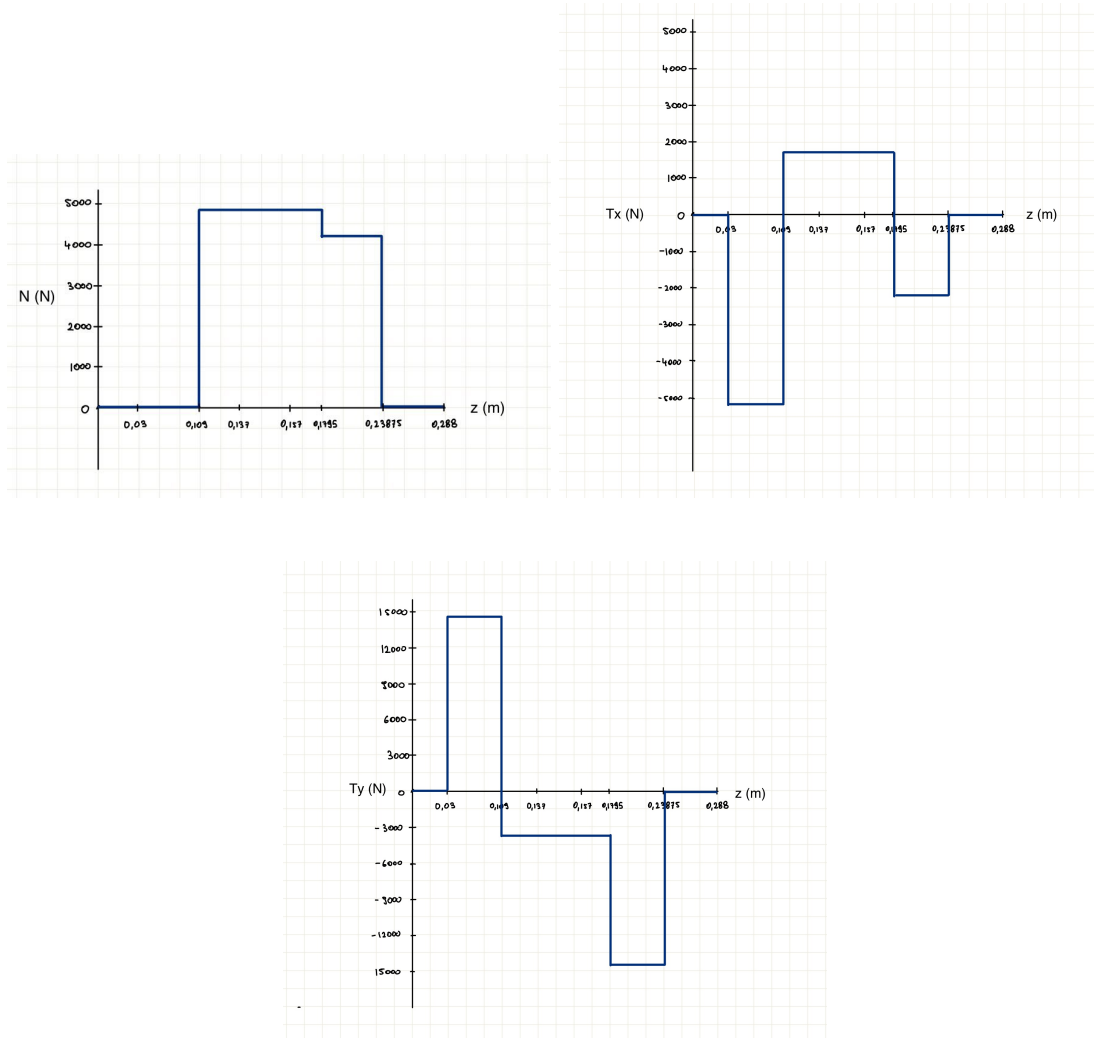


Figure 8

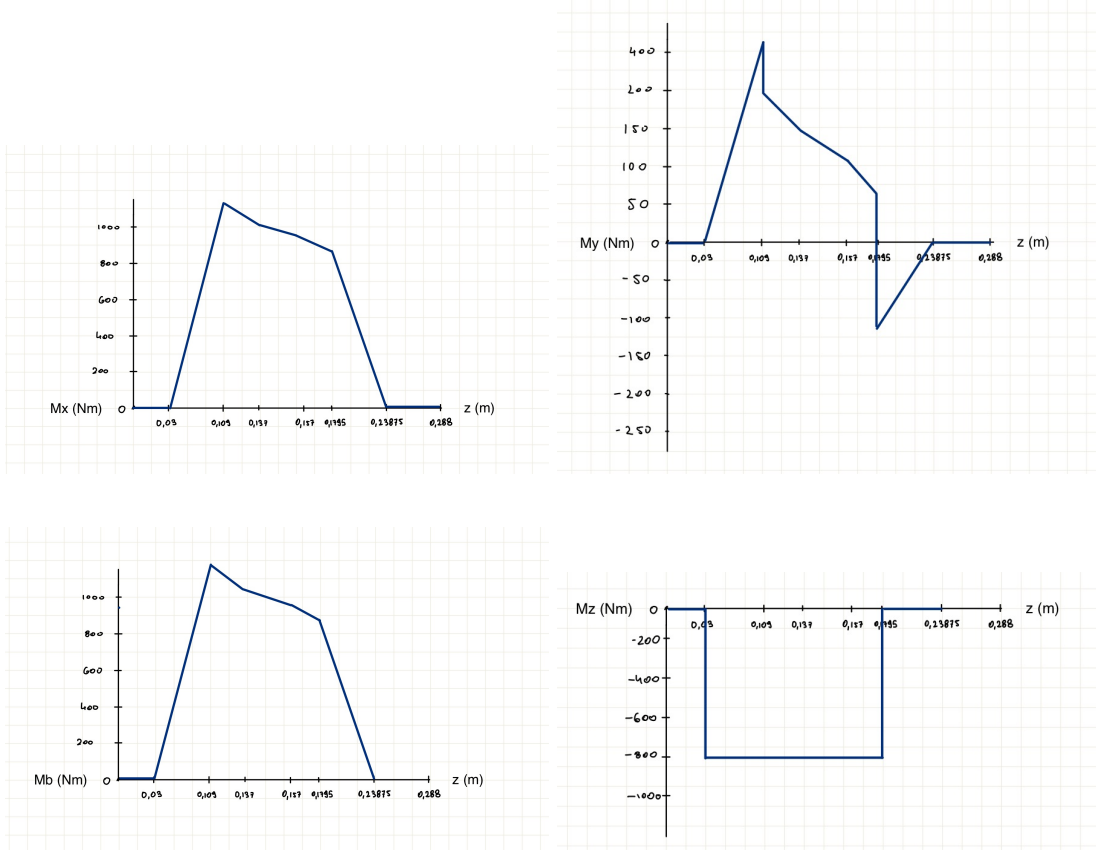


Figure 9

1.3.2 Stresses and static verification

Now the stresses can be calculated. The normal stress due to the normal force N can be simply obtained as:

$$\sigma_N = \frac{N}{A} \quad A = \frac{\pi d^2}{4} \text{ mm}^2$$

Likewise, the normal stress due to the bending moment can be calculated using the formula:

$$\sigma_{M_b} = \frac{M_b}{W_b} \quad W_b = \frac{\pi d^3}{32} \text{ mm}^3$$

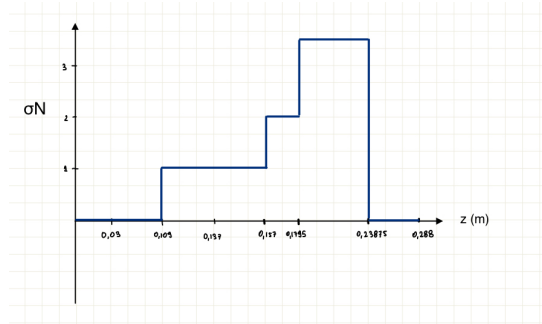
And lastly, the shear stress τ due to the torsional moment M_t can be attained using the formula:

$$\tau_{M_t} = \frac{M_t}{W_t} \quad W_t = \frac{\pi d^3}{16} mm^3$$

The formulas for the area A and the sectional moduli W_b and W_t are valid only for circular cross sections. It is also worth noting that since the cross-sectional diameter is not constant in different sections (values can be found in the technical drawing of shaft A2), the stresses will not depend on the force or moment only, but also on the position of the section as well (In different sections the shaft diameter changes). The values of the stresses are listed in the the following table. The trends of the normal stresses due to N and M_b and the shear stress due to M_t are

Section	d(mm)	σ_N (MPa)	σ_{M_b} (MPa)	τ_{M_t} (MPa)
C	45	0	0	0
V1	76.963	1.063	27.243	-8.9614
V2	62	1.64	44.79	-17.14
V3	56	2.007	56.004	-23.263
V4	42	3.568	121.149	-55.141
D	55	0	0	0

plotted in the graphs below:



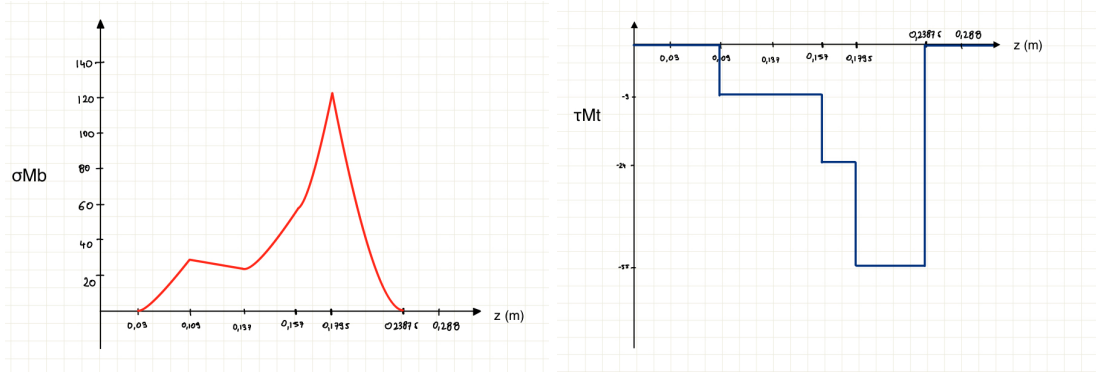


Figure 10

It does not come as a surprise that the shaft experiences high values of stress (particularly bending moment) in the V3 and V4 sections since these sections are located on gear 2 which is assembled on the shaft with a key while the gear 3 and the shaft are 1 piece. After calculating both contributions of the normal stress, the total normal stress can be simply found by summing them up:

$$\sigma_{tot} = \sigma_N + \sigma_{M_b}$$

The subsequent equivalent stress, useful for the calculation of the static safety factor can be obtained by considering the Von Mises criterion for full yielding.

$$\sigma_{eq} = \sqrt{\sigma_{tot}^2 + 3 \times \tau_{M_t}^2}$$

In the following graph, the trends of σ_{tot} and σ_{eq} are plotted together with the table of results:

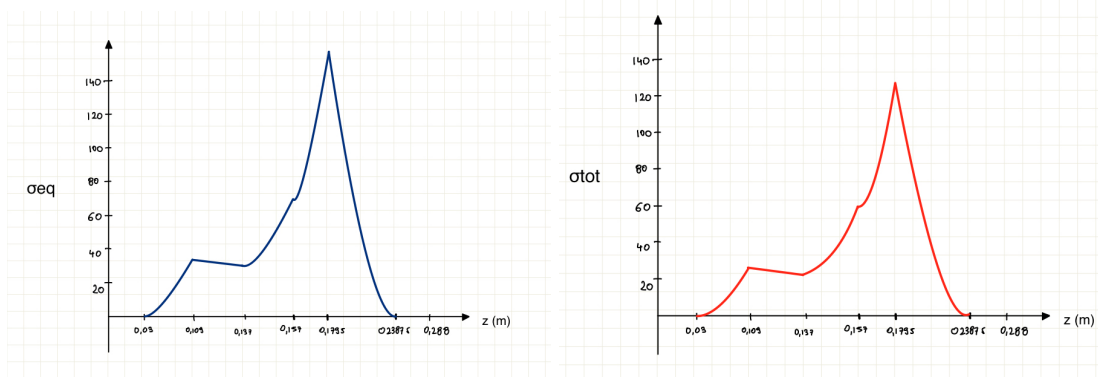


Figure 11

We can calculate the Safety Factor as:

$$SF = \frac{\sigma_y}{\sigma_{eq}}$$

Section	σ_{tot} (MPa)	σ_{eq} (MPa)	SF
V_1	28.305	32.28	28.809
V_2	46.42	55.11	16.87
V_3	58.01	70.631	13.167
V_4	124.72	157.086	5.92

From the given data $\sigma_y = 930$ MPa. The most critical section is section V4 since it has the lowest safety factor because of the presence of the key for assembling gear 2.

1.4 Fatigue verification

1.4.1 Evaluation of fatigue stresses

Fatigue verification is the process of testing the resistance of a component (or specimen) that is undergoing fluctuating stresses and strains. In addition, the number of cycles the component can endure at a certain load can be calculated to determine the component's fatigue life. To perform the fatigue test, the values of the mean and alternating stresses due to the normal force, bending moment, and torsional moment are required. Since in this situation, the torque of the shaft stays constant and the normal stress does not vary in time, the only load that, at each point of the cross-section changes with time due to the rotation of the shaft is the bending moment; this condition is also known as alternate bending. Therefore:

$$\sigma_m^N = \sigma_N \quad \sigma_a^N = 0$$

$$\sigma_m^{M_b} = 0 \quad \sigma_a^{M_b} = \sigma_{M_b}$$

$$\tau_m^{M_t} = \tau_{M_t} \quad \tau_a^{M_t} = 0$$

For the eventual calculation of the fatigue safety factors, the mean and alternate equivalent stresses are required. To obtain these values, as instructed, the Shigley method is used.

$$\sigma_{a,eq} = \sqrt{(K_f^{M_b} \times \sigma_a^{M_b} + \frac{K_f^N}{0.85} \times \sigma_a^N)^2 + 3 \times (K_f^\tau \times \tau_a^{M_t})^2}$$

$$\sigma_{m,eq} = \sqrt{(K_f^{M_b} \times \sigma_m^{M_b} + K_f^N \times \sigma_m^N)^2 + 3 \times (K_f^\tau \times \tau_a^{M_t})^2}$$

These formulas can be simplified because a few contributions are null. Therefore:

$$\sigma_{a,eq} = K_f^{M_b} \times \sigma_a^{M_b}$$

$$\sigma_{m,eq} = \sqrt{(K_f^N \times \sigma_m^N)^2 + 3 \times (K_f^\tau \times \tau_a^{M_t})^2}$$

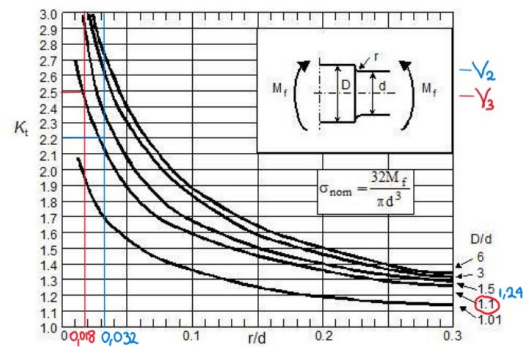
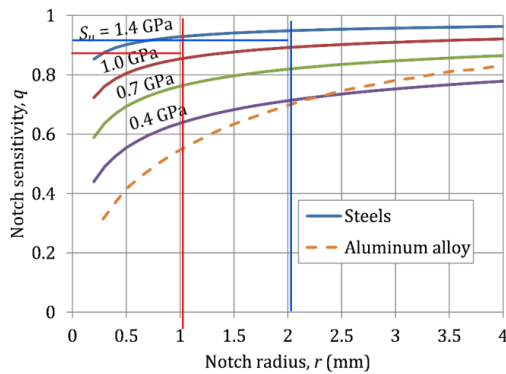
1.4.2 Stress concentration factors

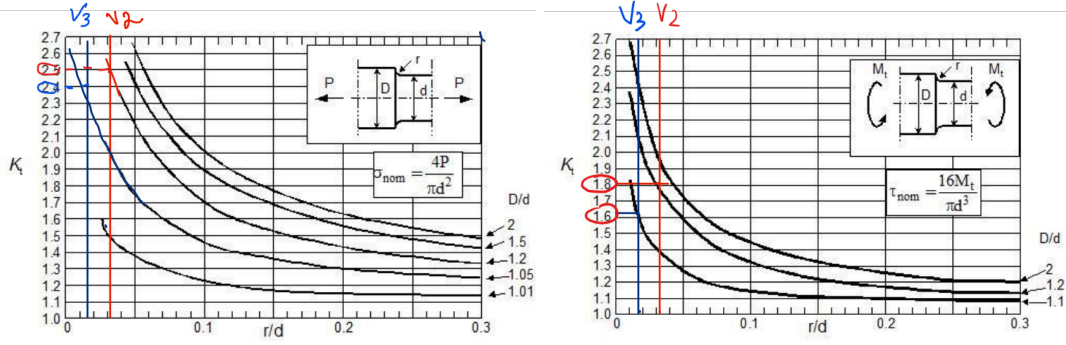
The study of the necessary factors can be carried out. It can be observed that there are two different situations in shaft A2. For section V4, there is a key-seat. In this case, the fatigue stress intensification factor is provided. It is assumed that for key-seats $K_f^N = 1$ and it is given that $K_f^{M_b} = 1.6$ and $K_f^{M_t} = 2$. On the other hand, for sections V2 and V3 we have a shoulder. K_f is calculated using the formula:

$$K_f = 1 + q \times (K_t - 1)$$

where q is the notch sensitivity factor and K_t is the geometrical stress concentration factor. The notch sensitivity factor q is a function of the ultimate strength σ_u of the material the shaft is made of and of the notch radius r of the shoulder. So $q(\sigma_u, r) = 0.91$ for V2 and $q(\sigma_u, r) = 0.88$ for V3. The values of K_t^N , $K_t^{M_b}$ and $K_t^{M_t}$ can be obtained from their respective graphs, all of which are presented below. In addition to the graphs, a table with the values of the factors and the stresses for all sections is provided.

Section	σ_N	K_t^N	K_f^N	σ_{M_b}	$K_t^{M_b}$	$K_f^{M_b}$	τ_{M_t}	$K_t^{M_t}$	$K_f^{M_t}$	q
V1	1.06	-	1	27.2	-	1.2	-8.96	-	1.5	0
V2	1.06	2.5	2.365	23.4	2.2	2.092	-8.96	1.8	1.728	0.91
V3	2	2.4	2.232	56	2.5	2.32	-23.3	1.6	1.528	0.88
V4	3.57	-	1	121.1	1	1.6	-55.1	-	2	0





1.4.3 Fatigue limit correction for the component and Haigh diagrams

The fatigue limit for the component σ_{D-1}^C can be corrected using the correction factors C_F (surface finish effect factor) and C_S (scale effect factor).

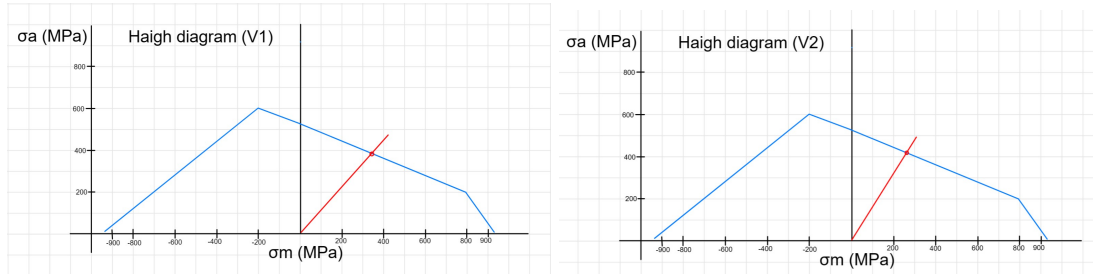
$$\sigma_{D-1}^C = C_F \times C_S \times \sigma_{D-1}$$

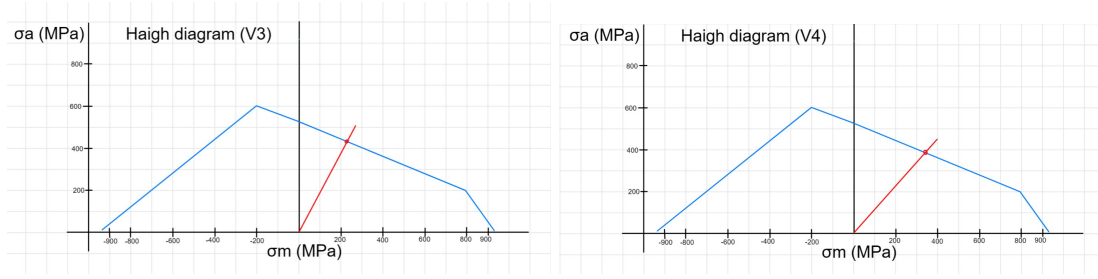
To find the safety factors, the limit stress σ_D^{lim} is needed and it can be found for all four cases as the intersection between the Haigh diagram and the respective working line. It's worth highlighting that both alternate and mean stresses are considered to vary with performances, which implies that the working lines will be oblique. When the latter intersects the Goodman line in the diagram the following formula has to be used:

$$\sigma_D^{lim} = \frac{\sigma_{D-1}^C}{1 + \frac{\sigma_m^{eq} \times \sigma_{D-1}^C}{\sigma_a^{eq} \times \sigma_u}}$$

If instead, a performance line crosses the yield strength σ_y line first, thus reaching the yield limit of the material, σ_D^{lim} will naturally belong to that line, otherwise, it would entail the yield strength limit would be exceeded, resulting in failure. The formula for σ_D^{lim} in this case becomes:

$$\sigma_D^{lim} = \frac{\sigma_y}{1 + \frac{\sigma_m^{eq}}{\sigma_a^{eq}}}$$





The Haigh diagrams for the four sections are added below. In addition, the values of the alternate and mean equivalent stresses and σ_D^{lim} can be found in the following table.

Section	σ_a^{eq}	σ_m^{eq}	σ_D^{lim}
V1	33	23	380
V2	49	27	420
V3	129.9	61.7	430
V4	193.8	191	370

Now the calculation of the fatigue safety factors can take place. Given this application, the proper definition of the fatigue safety factor would be:

$$SF_f = \frac{\sigma_D^{lim}}{\sigma_a^{eq}}$$

The values of the fatigue safety factors are:

Section	Safety factor
V1	11.76
V2	8.51
V3	3.34
V4	1.89

As for static verification, the lowest safety factor is the one at section V4. This means that out of all sections, fatigue failure is more likely to happen here

1.5 Gear tooth verification

For the purpose of bending and contact stress verification for gear 3 consider the following data, concerning the gear under study

$m_n(mm)$	Z	ψ	ϕ_n	Q_v	$w_{tooth}(mm)$	$\sigma_u(MPa)$	$\sigma_y(MPa)$	Hardness(HB)
4	21	15	20	8	56	1060	930	700

and the following forces, which are acting on the gear, as computed above:

$$F_{t43} = 18447.8N$$

1.5.1 Fatigue verification for bending

Let's start by the verification of the ability of the teeth to sustain bending cyclic stresses. To do so, the first step is computing the maximum bending stress acting on the gear tooth, in fatigue conditions, which is expressed by the following formula:

$$\sigma_{max,fatigue} = F_t K_0 K_B K_v K_H K_S \frac{1}{w_{tooth} m_t Y_j} = 242.17 MPa$$

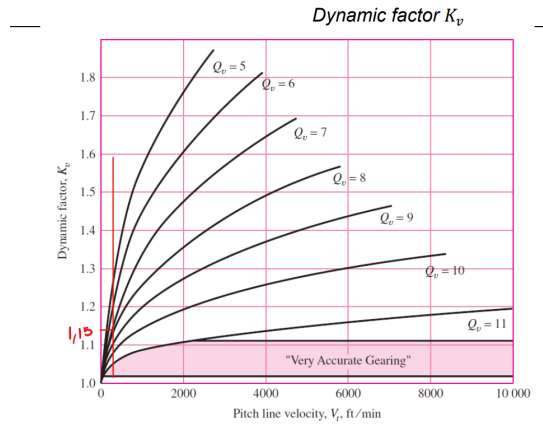
Obtained using the following factors:

- $K_0 = 1$ is the overload factor, for the considered condition of uniform power source and driven machine

Table of Overload Factors, K_o			
Driven Machine			
Power source	Uniform	Moderate shock	Heavy shock
Uniform	1.00	1.25	1.75
Light shock	1.25	1.50	2.00
Medium shock	1.50	1.75	2.25

• $K_B = 1$ is the rim-thickness factor and it is equal to 1 because gear 3 is mounted on shaft A2, so we don't have a rim.

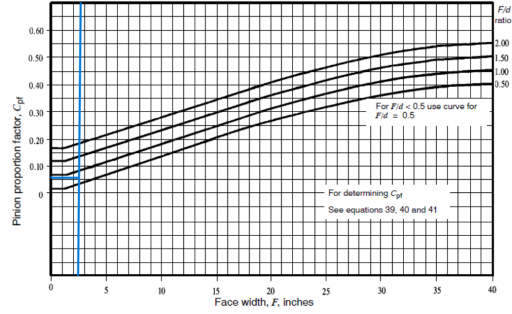
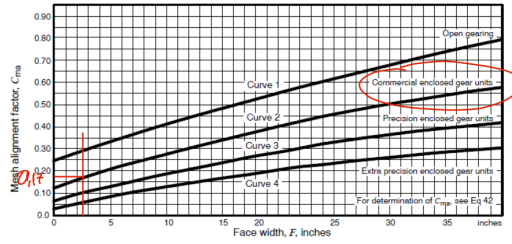
- $K_v = 1.15$ is the dynamic factor, considering
 - the accuracy transmission number $Q_v = 8$
 - the pitch line velocity $V_t = 286 \frac{ft}{min}$



- $K_H = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e)$

This is the load distribution factor computed considering:

- $C_{mc} = 1$ for uncrowned teeth
- $C_{pf} = 0.005$ ($S1/S = 0.12$ which is less than 0.175)
- $C_{pm} = 1$
- $C_e = 1$



- $K_s = 0.843(w_{tooth}m_t\sqrt{Y})^{0.0535} = 1.09$

$w_{tooth} = 56mm$ is the minimum face width between the one of the gear and the one of the pinion

$m_t = \frac{m_n}{\cos \psi} = 4.14mm$ is the transverse module of the the gear

$Y = 0.334$ obtained interpolating the highlighted data from the following table, considering for helical gears a virtual number of teeth given by :

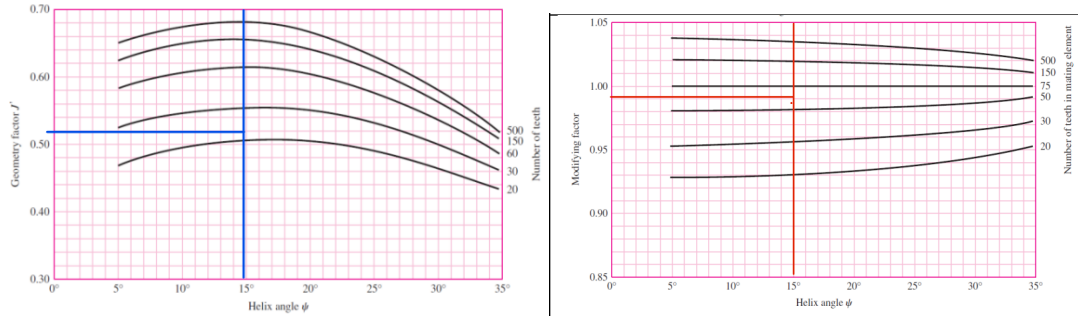
$$z' = \frac{z}{\cos^3 \psi}$$

Number of Teeth	Y	Number of Teeth	Y
12	0.245	28	0.353
13	0.261	30	0.359
14	0.277	34	0.371
15	0.290	38	0.384
16	0.296	43	0.397
17	0.303	50	0.409
18	0.309	60	0.422
19	0.314	75	0.435
20	0.322	100	0.447
21	0.328	150	0.460
22	0.331	300	0.472
24	0.337	400	0.480
26	0.346	Rack	0.485

- $Y_J = J'J'' = 0.5049$ is the bending strength geometry factor

$J' = 0.51$

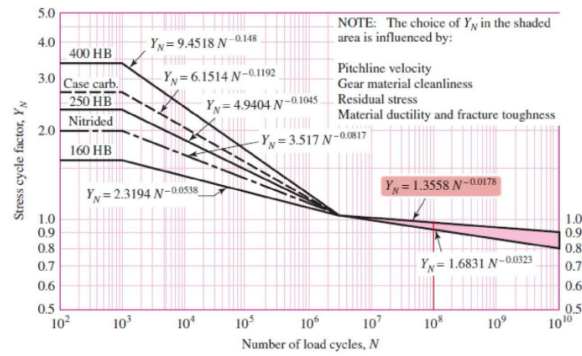
$$J'' = 0.99$$



Then the stress computed is used to evaluate the bending safety factor which is:

$$S_F = \frac{\sigma_{FP}}{\sigma_{max, Fatigue}} \frac{Y_N}{Y_\theta Y_Z}$$

- $\sigma_{FP} = 860 MPa$ is the bending fatigue strength
- $Y_N = 1.3558 N^{-0.0178} = 0.976 N$ is the stress cycle life factor, given $N = 10^8$



- $Y_\theta = 1$ is the temperature factor, as the working temperature is lower than $120^\circ C$
- $Y_Z = 1$ is the reliability factor, given a reliability equal to 0.99

Reliability	$K_R (Y_Z)$
0.9999	1.50
0.999	1.25
0.99	1.00
0.90	0.85
0.50	0.70

1.5.2 Fatigue verification for contact stresses

Now the verification for fatigue due to contact stress must be performed as well. As it was previously done the maximum pitting resistance of the gear tooth is computed, as shown in the following formula

$$\sigma_{max,Pitting} = Z_E \sqrt{F_t K_0 K_v K_S \frac{K_H Z_R}{w_{tooth} d_P Z_I}} = 1046.23 MPa$$

Besides the factors computed before, some additional ones are exploited:

- $Z_E = 191 \sqrt{MPa}$ is the elastic coefficient, considering both gears meshing as made of steel

Pinion Material	Pinion Modulus of Elasticity E_p (psi (MPa))*	Gear Material and Modulus of Elasticity E_G , lbf/in ² (MPa)*					
		Steel 30×10^6 (2×10^5)	Malleable Iron 25×10^6 (1.7×10^5)	Nodular Iron 24×10^6 (1.7×10^5)	Cast Iron 22×10^6 (1.5×10^5)	Aluminum Bronze 17.5×10^6 (1.2×10^5)	Tin Bronze 16×10^6 (1.1×10^5)
Steel	30×10^6 (2×10^5)	2300 (191)	2180 (181)	2160 (179)	2100 (174)	1950 (162)	1900 (158)
Malleable iron	25×10^6 (1.7×10^5)	2180 (181)	2090 (174)	2070 (172)	2020 (168)	1900 (158)	1850 (154)
Nodular iron	24×10^6 (1.7×10^5)	2160 (179)	2070 (172)	2050 (170)	2000 (166)	1880 (156)	1830 (152)
Cast iron	22×10^6 (1.5×10^5)	2100 (174)	2020 (168)	2000 (166)	1960 (163)	1850 (154)	1800 (149)
Aluminum bronze	17.5×10^6 (1.2×10^5)	1950 (162)	1900 (158)	1880 (156)	1850 (154)	1750 (145)	1700 (141)
Tin bronze	16×10^6 (1.1×10^5)	1900 (158)	1850 (154)	1830 (152)	1800 (149)	1700 (141)	1650 (137)

Poisson's ratio = 0.30.

$$Z_I = \frac{\cos \psi_t \sin \psi_y}{2m_N} \frac{m_G}{m_G + 1} = 0.194$$

This is the surface strength geometry, for external meshing, computed considering:

- $\psi_t = 20.64^\circ$ is the transverse pressure angle
- $m_G = \frac{z_P}{z_G} = \frac{21}{63}$ is the gear ratio
- $m_N = \frac{p_n \cos \phi_n}{0.95 Z} = 0.6376$

$$*Z = Z_A + Z_B - (r_P + r_G) \sin \phi_t = 19.49 mm$$

Given that :

- $r_P = 43.48 mm$
- $r_G = 130.44 mm$
- $r_{bp} = 40.69 mm$
- $r_{bg} = 122.06 mm$
- $a = m_n = 4 mm$

Z_A and Z_B are respectively:

$$Z_A = \min[\sqrt{(r_P + a)^2 - r_{bp}^2}, (r_P + r_G) \sin \phi_t] = 24.47 mm$$

$$Z_B = \min[\sqrt{(r_G + a)^2 - r_{bg}^2}, (r_P + r_G) \sin \phi_t] = 56.34 \text{ mm}$$

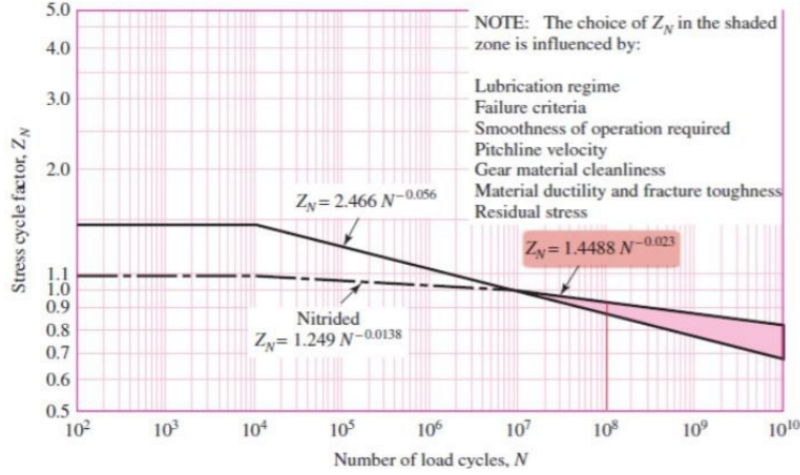
Similarly, the wear safety factor is then:

$$S_F = \frac{\sigma_{HP}}{\sigma_{max, Pitting}} \frac{Z_N Z_W}{Y_\theta Y_Z} = 1.33$$

This value is above the minimum one to be adopted, so the gear tooth is able to sustain pitting.

Again, a few more factors are needed, with respect to the bending fatigue case:

$-\sigma_{HP} = 1470 \text{ MPa}$ is the contact strength $-Z_N = 1.4488 N^{-0.023} = 0.948$ is the stress cycle life factor, given $N = 10^8$



$Z_W = 1 + A'(m_G - 1)$ is the hardness ratio factor

- $A' = 0$ as the two gears meshing have the same Brinell Hardness ($\frac{HB_P}{HB_G} < 1.2$)

- $m_G = \frac{z_P}{z_G} = \frac{21}{63}$ is the gear ratio

2 Part II

2.1 Bearings Data

From the reactions on the shafts we can get the forces on the bearings. Note that for bearing C the radial load is divided by two because there are two bearings ($15433.55 \text{ N}/2$):

Bearing data

Bearing	Fr (N)	Axial force (Fa) (N)	Shaft angular speed (rpm)
A	3400.248	–	1000
B	8121.158	2883.463128	1000
C	7716.774	–	333.333
D	14872.36	2059.61652	333.333
E	11726.68	–	111.111
F	20274.08	7349.502387	111.111

From the SKF catalogue we get the calculation factors, the minimum load factor, the reference speed and the limiting value:

Calculation factors

Bearing	Designation	X	Y	Y_0	Y_2	e	Reference Page
A	NU 206 ECP	-	0.4	-	-	0.3	509
B	30206 /DF	-	1.8	-	-	0.35	745
C	NU 209 ECP	-	0.6	-	-	0.3	509
D	32011 X/DF	0.5	1.2	0.9	0.6	0.44	745
E	NU 2210 ECP	-	0.8	-	-	0.3	509
F	32011 X/DF	0.5	1.2	0.9	0.6	0.44	745

Basic load ratings for bearings

Bearing	C_0 (kN)	C (kN)	P_u (kN)	k_r	n_r (rpm)	Reference
A	36.5	44	4.5	0.15	13500	518
B	88	85.7	9.65	-	7500	744
C	64	69.5	8.15	0.15	9000	520
D	232	170	26	-	4500	744
E	88	90	11.4	0.12	8500	522
F	232	170	26	-	4500	744

In the same pages of the SKF catalogue we can also find the bore and the outside diameters of the bearings, from which we can calculate the average diameter:

Diameters s

Bearing	d (mm)	D (mm)	d_m (mm)
A	30	62	46
B	30	62	46
C	45	85	65
D	55	90	72.5
E	50	90	70
F	55	90	72.5

2.2 Bearings Static Verification

2.2.1 Bearing minimum Load verification

Now we can calculate the minimum required radial force using the formulas:

$$F_m = k_r \left(6 + 4 \frac{n}{n_r} \right) \frac{d_m^2}{100}$$

Cylindrical Roller Bearings $\hookrightarrow$

$$F_m = 0.002C$$

Tapered Roller Bearings $\hookrightarrow$

Bearing	F_{rm} (kN)
A	0.2002062
B	1.714
C	0.4084167
D	3.4
E	0.6341176
F	3.4

The value of the radial loads applied on the bearings as reported in 10 are higher than the minimum required radial loads.

2.2.2 Static Safety Factor calculation

For calculating the static safety factor of the bearings we have to find the equivalent static load. From page 509 of the SKF catalogue we see that for the non-locating cylindrical roller bearings A, C, and E, we get that $P_0 = F_r$. While, for the locating tapered roller bearings B, D, and F we apply the following procedure as described at page 680 of the SKF catalogue:

$$P_0 = 0.5 * F_r + Y_0 * F_a$$

$$P_0 < F_r \rightarrow P_0 = F_r$$

values of $\frac{F_a}{F_r}$ and e

Bearing	$P_0 = 0.5 * F_r + Y_0 * F_a$ (N)	F_r (N)
B	9250.812593	8121.157928
D	10731.56471	14872.35656
F	21896.24248	20274.07733

We can summarize here the values of P_0 and the safety factor, calculated as C_0/P_0 for the bearings:

Basic load ratings for bearings

Bearing	C_0 (kN)	P_0 kN	SF
A	36.5	3400.25	10.735
B	88	9250.81	9.5127
C	64	15433.55	4.1468
D	232	14872.36	15.5993
E	88	11762.68	7.504
F	232	21896.24	10.595

2.3 Fatigue Verification

2.3.1 Equivalent Dynamic Load Calculation

For calculating the safety factor of the bearings we have to find the equivalent load. From page 509 of the SKF catalogue we see that for the non-locating cylindrical roller bearings A, C, and E, we get that $P = F_r$. While, for the locating tapered roller bearings B, D, F we have to

evaluate the $\frac{F_a}{F_r}$ and compare it with the calculation factor e as reported in page 680 of the SKF catalogue:

$$\frac{F_a}{F_r} \leq e \rightarrow P_u = F_r + Y_1 * F_a \quad \frac{F_a}{F_r} > e \rightarrow P_u = 0.67 * F_r + Y_2 * F_a$$

values of $\frac{F_a}{F_r}$ and e

Bearing	$\frac{F_r}{F_a}$	e	Formula used
B	0.355055665	0.37	$P_u = F_r + Y_1 * F_a$
D	0.138486225	0.40	$P_u = F_r + Y_1 * F_a$
F	0.362507367	0.40	$P_u = F_r + Y_1 * F_a$

2.3.2 Lubricant Choice

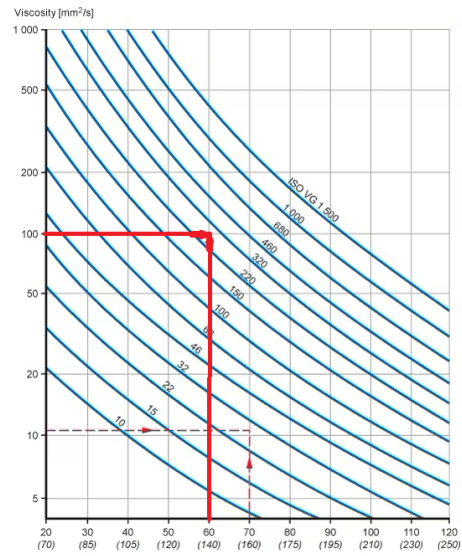
The viscosity is dependent on the value of the average radius d_m and that of the angular velocity n . The angular velocity of bearings is the same of that of the gears mounted or manufactured on the same shaft. In particular $n_A = n_B = n_{G1}$, $n_C = n_D = n_{G2} = n_{G3}$ and $n_E = n_F = n_{G4}$. These values are recollected for all the bearings in the following table.

values of $\frac{F_a}{F_r}$ and e

Bearing	d_m (mm)	n (rpm)
A	46	1000
B	46	1000
C	65	333.333
D	72.5	333.333
E	70	111.111
F	72.5	111.111

We select a lubricant so that we have a value of the viscosity ratio k as close as possible to one on the output shaft. From the graph we estimate first the rated viscosity on shaft A_3 .

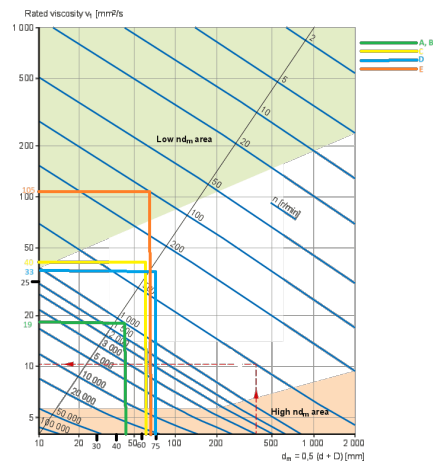
The rated viscosity for shaft A_3 is $100mm^2/s$, therefore we choose lubricant ISO VG 270 which, at $60^\circ C$ gives a viscosity of $100mm^2/s$.



Viscosity Diagrams for Bearing F

2.3.3 Rated Viscosity Evaluation

From the same graph we estimate the rated viscosity of the other bearings.



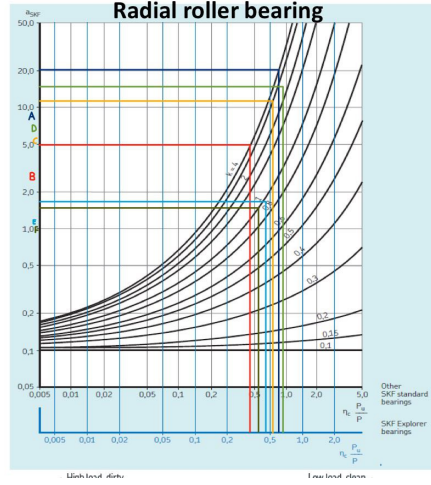
Rated Viscosity bearings A, B, C, D and E

Viscosity Ratio calculation

Bearing	$\nu_1 (mm/s)$	$\nu (mm/s)$	κ	$\eta \frac{P_u}{P}$	a_{skf}
A	100	19	5.26	0.661	20
B	100	19	5.26	0.362	5
C	100	40	3.03	0.528	11.5
D	100	33	2.5	0.707	15
E	100	105	0.95	0.486	1.7
F	100	100	1	0.397	1.6

2.3.4 Estimation of Bearings Life

Now we can estimate from the graph the value of a_{SKF} for each bearing.



Estimation of a_{SKF} for every bearing

Finally, we can calculate the life of the bearings both in number of cycles and in number of hours.

$$L_{m10} = a_1 a_{SKF} \frac{C^p}{P}$$

Number of Cycles $\hookrightarrow$

$$L_{m10h} = \frac{10^6}{60 * n} * L_{m10}$$

Life in Number of Hours $\hookrightarrow$

Bearing	L_{m10}	L_{mh10}
A	101742.7384	1695712.307
B	2482.149197	41369.15328
C	17479.45789	873972.982
D	24942.03557	1247101.903
E	1515.910395	227386.5819
F	386.7583646	58013.76049

Where we have considered a reliability equal to 90% ($a_1 = 1$) and have used $p = \frac{1}{3}$ for roller bearings.

3 Summary Tables

3.1 Shaft Analysis

	Value	Unit of Measurements
$R_{x,C}$	5228.96	N
$R_{y,C}$	14520.758	N
$R_{x,D}$	-2332.57	N
$R_{y,D}$	14688.298	N
$R_{z,C}$	0	N
$R_{z,D}$	-2059.62	N
M_x (z = V1)	1147.14	Nm
M_x (z = V2)	1037.18	Nm
M_x (z = V3)	958.64	Nm
M_x (z = V4)	870.28	Nm
M_y (z = V1)	198.15	Nm
M_y (z = V2)	149.92	Nm
M_y (z = V3)	115.48	Nm
M_y (z = V4)	-138.205	Nm
N (z = V1)	4943.08	N
N (z = V2)	4943.08	N
N (z = V3)	4943.08	N
N (z = V4)	2059.62	N
$Mb_{tot,max}$	1164.128	Nm
$Mt_{tot,max}$	-802.14	Nm
SF (static)	5.92	-
$\sigma_{a,b,nom}$ (z=m.c.s)	121.1	MPa
$\sigma_{m,b,nom}$ (z=m.c.s)	0	MPa
$\sigma_{a,N,nom}$ (z=m.c.s)	0	MPa
$\sigma_{m,N,nom}$ (z=m.c.s)	3.57	MPa

	Value	Unit of Measurements
$\tau_{a,t,nom}$ (z=m.c.s)	0	MPa
$\tau_{m,t,nom}$ (z=m.c.s)	-55.1	MPa
$K_{f,b}$	1.6	-
$\sigma_{a,b,eff}$ (z=m.c.s)	193.76	MPa
$\sigma_{m,b,eff}$ (z=m.c.s)	0	MPa
$K_{f,N}$	1	-
$K_{f,T}$	2	-
$\tau_{a,t,eff}$ (z=m.c.s)	0	MPa
$\tau_{m,t,eff}$ (z=m.c.s)	-110.2	MPa
$\sigma_{a,eqvP}$ (z=m.c.s)	193.8	MPa
$\sigma_{m,eqvP}$ (z=m.c.s)	191	MPa
SF_f (fatigue)	1.89	-

3.2 Gear Analysis

3.2.1 Bending

	Value	Unit of Measurement
$\frac{F_t}{b \times m_t \times Y_j}$	157.562	MPa
$\sigma_{max,fatigue}$	242.17	MPa
S_F	3.468	-

3.2.2 Pitting

	Value	Unit of Measurement
$Z_E \times \sqrt{\frac{F_t \times}{b \times d_p \times Z_I}}$	872.37	MPa
$\sigma_{max,pitting}$	1046.23	MPa
S_F	1.33	-

3.3 Bearing Analysis

3.3.1 Load Caluclation

	Value	Unit of Measurement
$F_{r,A}$	3.4	kN
$F_{a,A}$	0	kN
$F_{r,B}$	8.12	kN
$F_{a,B}$	2.88	kN
P_A	3.4	kN
P_B	13.31	kN
$F_{r,C}$	7.71	kN
$F_{a,C}$	0	kN
$F_{r,D}$	14.87	kN
$F_{a,D}$	2.059	kN
P_C	7.71	kN
P_D	18.37	kN
$F_{r,E}$	11.73	kN
$F_{a,E}$	0	kN
$F_{r,F}$	20.27	kN
$F_{a,F}$	7.35	kN
P_E	11.72	kN
P_F	32.77	kN

3.3.2 Corrected Rating life

	Value	Unit of Measurement
ISO VG of the selected oil	220	-
$a_{skf,A}$	20	-
$a_{skf,B}$	5	-
$a_{skf,C}$	11.5	-
$a_{skf,D}$	15	-
$a_{skf,E}$	1.7	-
$a_{skf,F}$	1.6	-
$L_{10mh,A}$	1695712	Hours
$L_{10mh,B}$	41369.15	Hours
$L_{10mh,C}$	873973	Hours
$L_{10mh,D}$	1247101.9	Hours
$L_{10mh,E}$	227386.582	Hours
$L_{10mh,F}$	58013.7605	Hours